

Team Name: Three Amigos

Project Proposal

ECE 470: Artificial Intelligence

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May 25, 2026

Problem Statement

Inflation is taking people's purchasing power by around 3-8% per year since 2022, while banking investments/savings accounts give them around 3-5% interest. Traditional banking is letting people down and draining their savings. Stock investments are a new way for people to not only maintain but also increase their purchasing power year by year. While investing in stocks can be risky, there are ways to mitigate the risk and maximize results. This project addresses this issue by using a genetic algorithm to optimize portfolio investments with the goal of determining the optimal allocation of funds among different stocks. In this case, the use of AI and the generic algorithm is a requirement due to the high density of data and the complexity of marker dynamics in order to make accurate predictions.

Problem Formulation

The objective of the project is to maximize portfolio performance. To that end, the Sharpe Ratio will be used as the fitness function:

$$Ratio = (Portfolio\ return - risk\ free\ returns) \div portfolio\ volatility$$

The search space for this problem is the possible allocation of funds across the stocks in the portfolio, adding up to 100% of all funds allocated. The size of this search space would be the number of possible portfolios that can be created by splitting up 100% into X number of stocks, avoiding fractional percentages in order to deal with the problem of dealing with infinite possibilities.

Programming Language

The project will utilize the Python programming language in conjunction with the following libraries:

- NumPy
- Pandas
- Matplotlib

Other libraries will be included as required by the project.

Evaluation Approach

In order to evaluate the model, we will compare the generated portfolios against existing portfolios, randomly assigned portfolios, and a simple division of portfolios. The dataset that will be used is 5-50 stocks selected from freely available financial datasets such as Yahoo Finance, Google Finance, or the S&P 500.

Milestone

The following are expected stages for the project:

- Select a list of stocks to include, download, and clean the dataset by June 12th (Santiago Velasco)
- Genetics Algorithm barebone by June 30th (Victor Mendes & Daniel Alvarez)
- Progress Report Draft by June 25th (everyone)
- Poster by July 14th (everyone)

References

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